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Graded Assignment • 20 min

Due Jul 24, 11:59 PM IST

Your grade: 100%

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1. You flip a fair coin 3 times. What is the probability of getting at least one head? 1 / 1 point

- ○ $\frac{3}{8}$
- ○ $\frac{1}{8}$
- ○ $\frac{5}{8}$
- ◉ $\frac{7}{8}$

Nice work

The complement of at least one head is all tails, with probability $(0.5)^3 = \frac{1}{8}$. Thus, $P(\text{at least one head}) = 1 - \frac{1}{8} = \frac{7}{8}$.

2. Consider throwing a biased die with probability of each side is dependent to its value with the following relation: $P(1) = \frac{1}{2}P(2) = \frac{1}{3}P(3) = \frac{1}{4}P(4) = \frac{1}{5}P(5) = \frac{1}{6}P(6)$. 1 / 1 point

What is the probability of seeing a prime number?

- ◉ $\frac{10}{21}$
- ○ $\frac{13}{21}$
- ○ $\frac{11}{21}$
- ○ $\frac{12}{21}$

Nice work

Note that $P(1) + P(2) + \cdots + P(6) = 1 \implies 21P(1) = 1 \implies P(1) = \frac{1}{21}$. The probability of others sides can be easily calculated. We must see a prime. Which means we need to calculate $P(2) + P(3) + P(5) = \frac{2}{21} + \frac{3}{21} + \frac{5}{21} = \frac{10}{21}$

3. Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. What is $P(A \cap B)$ and $P(A \cup B)$? 1 / 1 point

- ○ $0.02, 0.7$
- ◉ $0.2, 0.7$
- ○ $0.6, 0.9$
- ○ $0.1, 0.9$

Nice work

For independent events, $P(A \cap B) = P(A) \cdot P(B) = 0.4 \cdot 0.5 = 0.2$. $P(A \cup B) = P(A) + P(B) – P(A \cap B) = 0.7$.

4. Assume we generate a 6-digit number using the following digits $1, 2, 3, 4, 5, 6$ such that all digits are different. What is the probability that no odd digits are next to each other? 1 / 1 point

- ◉ $\frac{1}{5}$
- ○ $\frac{1}{4}$
- ○ $\frac{4}{15}$
- ○ $\frac{2}{5}$

Nice work

$\frac{\binom{4}{3} \times 3! \times 3!}{6!} = \frac{1}{5}$.

5. A bag contains 3 red balls and 2 blue balls. You draw one ball and place it in a second bag with 1 red and one blue ball. Then, you draw one ball from the second bag. What is the probability that it is red? 1 / 1 point

- ○ $\frac{7}{15}$
- ◉ $\frac{8}{15}$
- ○ $\frac{3}{5}$
- ○ $\frac{1}{3}$

Nice work

The probability of drawing a red ball from the first bag is $\frac{3}{5}$, making the second bag have 2 red and 1 blue (probability red = $\frac{2}{3}$). If a blue ball is drawn ($\frac{2}{5}$), the second bag has 1 red and 2 blue (probability red = $\frac{1}{3}$). Total probability: $\frac{3}{5} \cdot \frac{2}{3} + \frac{2}{5} \cdot \frac{1}{3} = \frac{6}{15} + \frac{2}{15} = \frac{8}{15}$.

6. Assume there are 3 people from Hong Kong, 4 people from France and 5 people from Spain. We select a team of 3 people at random. Answer the following questions respectively from left: 1 / 1 point

(i) What is the probability that they are all from different countries?

(ii) What is the probability that they are all from same country?

(iii) What is the probability that exactly two of them are spanish?

- ◉ $\frac{3}{11}, \frac{3}{44}, \frac{7}{22}$
- ○ $\frac{2}{11}, \frac{1}{11}, \frac{1}{22}$
- ○ $\frac{6}{11}, \frac{5}{11}, \frac{1}{11}$
- ○ $\frac{7}{44}, \frac{2}{11}, \frac{7}{22}$

Nice work

There are $\binom{12}{3}$ different ways to pick 3 random people.

1. $\frac{\binom{3}{1} \times \binom{4}{1} \times \binom{5}{1}}{\binom{12}{3}}$
2. $\frac{\binom{3}{3} + \binom{4}{3} + \binom{5}{3}}{\binom{12}{3}}$
3. $\frac{\binom{5}{2} \cdot \binom{7}{1}}{\binom{12}{3}}$

7. Assume you draw one card from a standard 52-card deck uniformly at random. What is the probability that it is a heart? 1 / 1 point

- ○ $\frac{1}{2}$
- ◉ $\frac{1}{4}$
- ○ $\frac{1}{52}$
- ○ $\frac{1}{13}$

Nice work

There are 13 hearts in a 52-card deck, so the probability is $\frac{13}{52} = \frac{1}{4}$.

8. You roll two fair six-sided dice. What is the probability that the sum is 7? 1 / 1 point

- ○ $\frac{1}{8}$
- ○ $\frac{1}{3}$
- ○ $\frac{2}{9}$
- ◉ $\frac{1}{6}$

Nice work

There are 6 outcomes for sum 7out of 36 possible outcomes, so $P(\text{sum } 7) = \frac{6}{36} = \frac{1}{6}$.